

## Chomsky Hierarchy:-

The class of phrase structure grammars is very large but, we can put or impose certain constraints on the production rules.

This leads to different classes of phrase structure grammar depending on different restrictions.

Chomsky had suggested four different classes of phrase structure grammars as follows:-

- 1) Type 0 Un-restricted grammar
- 2) Type 1 Context sensitive grammar
- 3) Type 2 Context free grammar
- 4) Type 3 Regular grammar.

### 1) Unrestricted grammar - Type 0

a) There is no restrictions on the productions of grammar of this type.

2) This type of grammar permits -

$$\alpha \rightarrow \beta \quad \text{with } \alpha \neq \epsilon$$

where

$\alpha \neq \epsilon$   $\beta$  are sentential forms

(Any combination of terminals or non-terminals)

$$\text{i.e. } \alpha, \beta \in (V \cup T)^*$$

Such a grammar is called as Unrestricted grammar, semi-Thur grammar.

3) This grammar generates the recursively enumerable language.  
or every type 0 - forms recursive enumerable language.

4) We can construct Turing m/c to recognize the sentences generated by this type of grammar.

e.g

Let

$$V = \{A, B, C\}$$

$$T = \{a, b, c\}$$

$$G = \{V, T, P, A\}$$

where  $P$  consist of,

$$A \rightarrow AB$$

$$AB \rightarrow AC$$

$$B \rightarrow a$$

No restriction on productions

2) Context Sensitive Grammar - Type 1.

i) There is a restriction on this type of grammar are -

1) for each production of the form  $\alpha \rightarrow \beta$  where

length of  $\beta$  ( $|B|$ )  $\geq$  length of  $\alpha$  ( $|A|$ )

i.e  $\beta$  is atleast as much as the  $\alpha$

ii) The rule of the form  $A \rightarrow \epsilon$

is not allowed unless  $A$  is a

Start symbol. It does not occur on right hand side of any rule.

2) This grammar generates the context sensitive language

3) The Turing m/c can be constructed to recognize the context-sensitive language generated by Context Sensitive grammar (CSG)

4) e.g

$$S \rightarrow ABC$$

$$S \rightarrow SABC$$

$$aA \rightarrow aa$$

$$bC \rightarrow bc$$

$$cC \rightarrow cc$$

3) Context Free Grammar - Type 2.

1) In this grammar, allowed type of production is,

$$A \rightarrow \alpha$$

where  $A$  is a nonterminal &  $\alpha$  is a sentential form i.e.  $\alpha \in (V \cup T)^*$

(Supposingly  $A$  can be equal to  $\epsilon$ )

2) Left hand side of production should contain only one non-terminal.

3) Start symbol can appear on right hand side also.

3) This grammar generates context-free languages (CFL) [generated by CFG]

- 4) PDA push down automata can accept CFL.

e.g

Let Grammar  $G$  is,

$$G = (\{S\}, \{a, b\}, P, S)$$

where  $P$  consists of the following productions,

$$S \rightarrow aSa \mid bSb \mid a \mid b$$

- 4) Regular Grammar - Type 3.

1) In this type of productions there are restriction on the type of productions:-

- i) Left hand side of each production should contain only one nonterminal.
- ii) Right hand side can contain at the most one nonterminal which is allowed to appear as rightmost or leftmost symbol.

2) This grammar generates using this grammar regular languages

3) Finite state m/c recognizes regular languages. (Expressed using RE)

4) Depending on the type of nonterminal this type of grammar is further classified as-

- a) left linear grammar
- b) Right linear grammar

a) left linear grammar

\*) A regular grammar can contain at the most one nonterminal on the righthand side of its production.

\*) If this nonterminal appears as the leftmost symbol on the right hand side, the regular grammar is said to be left linear.

\*) Allowed type of productions in left linear grammar is, are,

$$A \rightarrow BW, A \rightarrow E \text{ or } A \rightarrow w$$

where  $A$  &  $B$  are variables &  $w$  is a string of terminals.

e.g

$$G = (\{S, B, C\}, \{a, b\}, P, S)$$

where  $P$  consist of

$$S \rightarrow Ca|Bb$$

$$C \rightarrow Bb$$

$$B \rightarrow Ba|b$$

b) Right-linear Grammar -

\* A Regular grammar can contain at the most one nonterminal on the right hand side and as rightmost symbol on the right hand side of the production then the grammar is said to be right linear grammar.

\* Allowed forms of the production are,

$A \rightarrow WB$ ,  $A \rightarrow w$  or  $A \rightarrow \epsilon$

where  $A$  &  $B$  are nonterminals and  $w$  is the string of terminals.

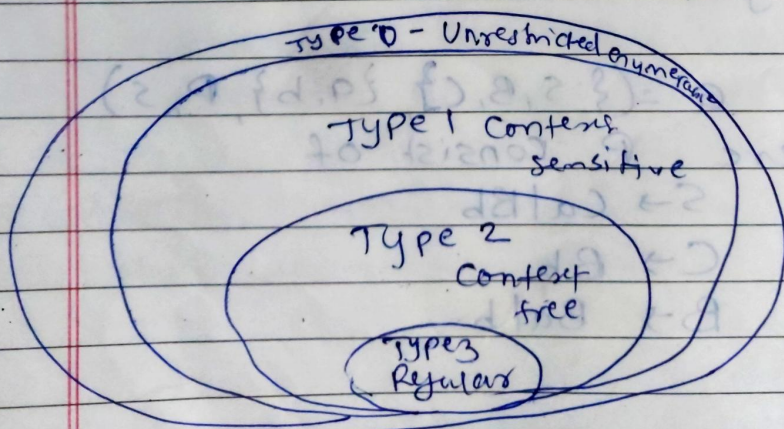
e.g

$G = (\{S, A\}, \{0, 1, \epsilon\}, P, S)$

where  $P$  consists of,

~~$S \rightarrow 0A$~~   $S \rightarrow 0A$

$A \rightarrow 0A | \epsilon$



# Summary of Chomsky Hierarchy:-

| language class | language                                   | Grammar                   | Machines                | Example       |
|----------------|--------------------------------------------|---------------------------|-------------------------|---------------|
| Type 3         | Regular                                    | Regular Grammar           | Fsm -<br>NFA/DFA        | $a^*b^*$      |
| Type 2         | Context Free                               | Context Free Grammar      | PDA                     | $a^n b^n$     |
| Type 1         | Decidable language                         | Context Sensitive Grammar | Linear Bounded Automata | $a^n b^n c^n$ |
| Type 0         | Computable language<br>(unrestricted lang) | Unrestricted Grammar      | Turing machine          | $n!$          |